LAB 6

Code and Branch Coverage Analysis

Bookstore Trade-In Application

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Course	Software Engineering
Repository	github.com/Darshil085/L6_L7_09_bookstore_trade_in

1. Objective

The objective of this lab was to use automated tests and the Coverage.py tool to measure application code coverage and branch coverage, identify untested execution paths, and improve the test suite to meet the required minimum of 90% for both metrics.

2. Application Under Test

The application module, **bookstore_trade_in.py**, contains two functions:

process_book_buybacks(batches, store_credit_bonus) — processes approved book buyback batches, evaluates book format and condition, rejects books with missing pages or counterfeit flags, applies store-credit bonuses, and returns the payout and processing status.

inspect_scanner_calibration(scanner_matrix) — evaluates scanner error readings and returns NO_DATA, CALIBRATED, or RECALIBRATE depending on the readings and consecutive error streaks.

3. Tools and Method

Python 3.13.7, pytest 9.2.0, and Coverage.py 7.17.0 were used. The initial test file was run to establish a baseline. Additional tests were then written in two separate files, one for code coverage and one for branch coverage. Coverage data was reset between the two runs so that each report measured its corresponding test file independently.

Commands used

```
python -m coverage run --source=bookstore_trade_in -m pytest
CodeCoverageTest_PES1UG25CS820_application.py
python -m coverage report -m

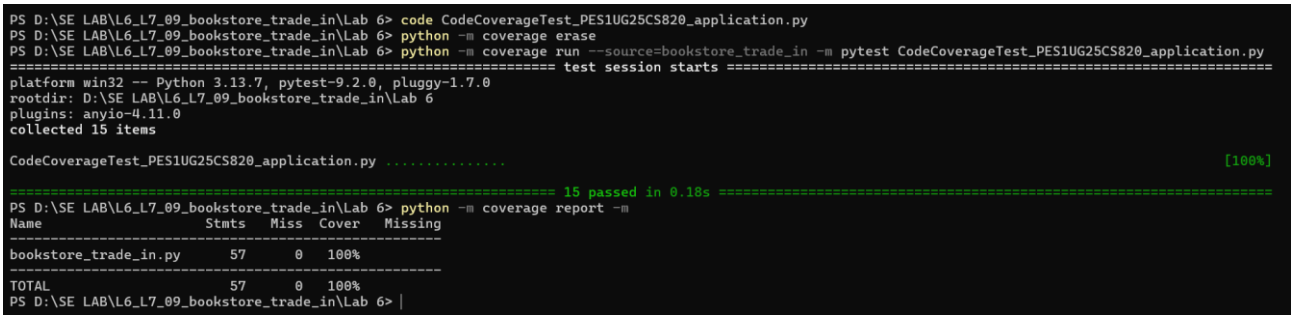
python -m coverage run --branch --source=bookstore_trade_in -m pytest
BranchCoverageTest_PES1UG25CS820_application.py
python -m coverage report -m
```

4. Results

100%	100%
Code / statement coverage	Branch coverage

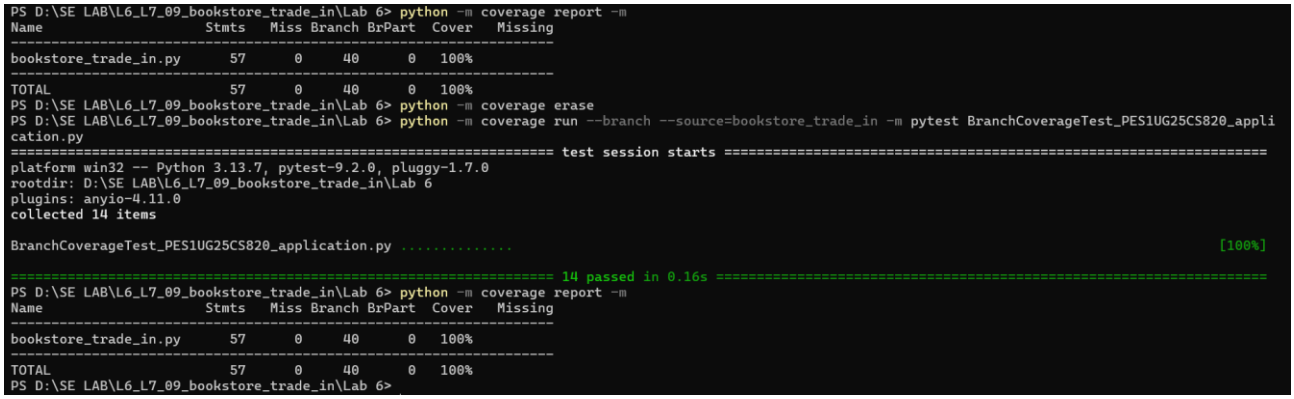
Metric	Final result	Required minimum	Outcome
Code coverage	57/57 statements; 0 missed	90%	PASS
Branch coverage	40 branches; 0 partial	90%	PASS
Code coverage test run	15 passed	—	PASS
Branch coverage test run	14 passed	—	PASS

Code coverage: 15 tests passed. Coverage.py reported 57 statements, 0 missed, and 100% coverage for bookstore_trade_in.py.



Branch coverage: 14 tests passed. Coverage.py reported 57 statements, 0 missed, 40 branches, 0 partial branches, and 100% coverage.

The initial baseline run had one original test passing and reported 49% coverage for the application module. The expanded tests exercised the previously uncovered statements and decision outcomes.



5. Test Scenarios

Buyback processing: invalid input and negative bonus; unapproved intake; textbook, hardcover, paperback, and unsupported formats; mint, poor, default, and other conditions; missing pages; counterfeit books including textbook

batch termination; store-credit bonus; audit rejection; zero-credit and completed payout statuses.

Scanner calibration: empty input; normal readings; error readings above the threshold; consecutive error detection; reset of a misread streak after an acceptable reading; and recalibration across multiple devices.

6. Version Control and Deliverables

The two test stages were committed separately, as required. Both commits were pushed successfully to the main branch of the individual repository.

Commit	Message	Files
6ee1040	Add code coverage test cases and screenshot	Code coverage test file + PNG
50a06fe	Add branch coverage test cases and screenshot	Branch coverage test file + PNG

Individual GitHub repository:
https://github.com/Darshil085/L6_L7_09_bookstore_trade_in

Lab 6 folder:
https://github.com/Darshil085/L6_L7_09_bookstore_trade_in/tree/main/Lab%206

Committed evidence screenshots: CodeCoverage_PES1UG25CS820.png and BranchCoverage_PES1UG25CS820.png. Both screenshots are stored in the Lab 6 folder in the repository and show the final terminal coverage results.

7. Conclusion

The expanded test suites achieved 100% code coverage and 100% branch coverage for the bookstore trade-in application. All 15 code coverage tests and all 14 branch coverage tests passed. The two test files and their corresponding terminal screenshots were committed in separate commits and pushed to the individual GitHub repository.

End of Report